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Beyond Accuracy: Clinical Safety, Explainability, Cross Validation, Statistical Significance and Computational Efficiency based Deep Learning Benchmark for Pneumonia Detection

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ABSTRACT Pneumonia is one of the main diseases causing adult and child mortality worldwide. The delay in proper diagnosis and lack of healthcare resources is one of the major contributors. Hence, this research proposed a large-scale deep learning benchmark that evaluates the clinical safety, explainability, cross-validity, accuracy, statistical significance, and computational efficient computer-aided diagnosis (CAD) system for adult and pediatric pneumonia detection. We proposed a clinically safe benchmark framework comparing 10 architectures of Convolutional Neural Networks and Vision Transformers with stratified 5-fold cross-validation and McNemar's test for deeper performance evaluation of deep learning models. This research study demonstrated that VGG-19 from Convolutional Neural Networks (CNNs) and Swin-Tiny from Vision Transformers (ViTs) exhibited majority-class prediction collapse in multiple folds and rendering them clinically unsafe. This research also experimentally confirms through McNemar testing that the EfficientNet-B0 significantly outperformed from all competing architectures. This research highlights that the computational efficiency of ResNet-18 model enables real-time screening in low-resource settings while EfficientNet-B0 provided the best accuracy-resource tradeoff. The implementation of explainability through Grad-CAM++ confirms that models focus on lung parenchyma and consistent with radiological best practices. The identification of the majority-class collapse issue in deep learning models provides actionable safety guidance for clinical deployment.

INDEX TERMS Pediatric Pneumonia, Adult Pneumonia, Chest X-Ray, CNNs, EfficientNet-B0, Vision Transformers, Grad-CAM++, McNemar's Test, Deep Learning, Medical Image Analysis

I. INTRODUCTION

PNEUMONIA is one of the major diseases related with lung infection, which leads to the filling of pus fluid and causes highest number of deaths worldwide. According to World Health Organization's (WHO) approximately 800,000 people die from pneumonia per year, 15% death ratio of children under five years of age. The pneumonia causes more deaths in low and middle-income countries, and imposing

more burden on health care system, specially, where lack of access to digital diagnostic technological lab facilities and shortage of well-trained radiologists [1], [2]. Healthcare system has made significant advancements in recent years in terms of technology utilization, yet pneumonia remain as a key public health issue for adult and children across the world. Delay in diagnosis and lack of proper treatment of pneumonia causes patient's death. This issue is creating complications for countries with limited healthcare resources and shortage of trained radiologist staff and digital diagnostic infrastructure for effective clinical decision making [3]. The primary method to diagnose pneumonia is Chest Radiography (CXR) due to its availability, low cost, and minimal emission of radiation. However, the results of chest radiography face various challenges, including image quality variations, matching with other pulmonary disease symptoms, and the diagnostic variability of adults and children. There is difference in the diagnostic test of pneumonia for children and adult, because of inter-radiologist disagreement and low dose radiations for children, which cause low image contrast with combined probability of noise, so in results, the diagnosis results of CXR varies specially for children under age 3 years, which is a worse and serious challenge in interpretation for pediatric chest radiography of children. The low dose radiations in children's chest radiography settings causes more noise, which creates more hurdles in critical structure details of chest radiograph images [4], [5].

The advancement of technology and artificial intelligence has significantly improved computer aided diagnosis (CAD) system and contributing in medical image diagnosis analysis. Deep learning models specially Convolutional Neural Networks (CNNs) are performing well through extracting and recognizing from chest x-ray images for pneumonia detection [6]. After the public availability of chest x-ray images datasets including the chestX-ray 14, a labelled dataset on 14 diseases and pediatric chest X-ray accelerated more attention in automatic pneumonia detection systems [7], [8]. The modern architectures including DenseNet, ResNet, and EfficientNet deep learning models have performed well and enhanced the pneumonia diagnostic results because of their innovative skip connections in the medical images dataset, feature learning, improving parameter efficiency, and removing the vanishing gradient impact within deep networks; specifically, the EfficientNet improved its performance through the use of a principled compound scaling technique, which contribute in balancing network width, depth, and input resolutions, obtaining higher accuracy resource trade-off compared with previous CNN architectures [9], [10].

The rise of Vision Transformers (ViTs) in recent studies has become a more powerful alternate of Convolutional Neural Networks (CNNs) Models and competing with the long-term control and supremacy of CNN architectures and rely on local receptive fields. The Vision Transformers (ViTs) use self-attention mechanism which makes model long-range dependencies and global contextual bond within images [11]. Following the performance of transformers, a Swin Transformer architecture has achieved good results in segmentation through semantics and object detection tasks. This Swin transformer worked on the attention mechanism applied through feature mapping from images and moving window logic for dense prediction tasks, while preserving the linear computational complexity regarding the size of the input image [12]. Nowadays the vision transformers are used for image classification task in the medical domain including the disease classification, which enhances radiological decision support.

Though the CNNs and Vision Transformers have given satisfied performance independently on chest X-ray analysis and diagnosis, still multiple gaps exist; as some previous research studies focused on CNNs or transformers models by evaluating limited number of architectures. Secondly, many research studies were conducted only on pneumonia classification accuracy without considering, clinical safety, computational efficiency or statistical importance. Thirdly, very few research studies performed the failure mode analysis of these models on pneumonia classification and highlighting the majority class collapse issue and clinically unsafety of these models. Lastly, the use of mixed large-scale benchmarking of CNN and vision transformers and their statistical significance on combined dataset of pediatric and adult remain scarce. Subsequently, there is a lack of rigorous benchmarking studies that jointly evaluate diagnostic performance, statistical significance validation, clinical safety, failure mode analysis, explainability, consistency and deployment efficiency under a unified experimental protocol using a large-scale mixed pediatric-adult dataset.

A. RESEARCH CONTRIBUTIONS

This study addresses all these limitations through comprehensive benchmarking. Our primary contributions are:

- We present a clinically safe comprehensive large-scale benchmark of comparing seven CNNs and three Vision Transformers architectures on a combined pediatric and adult chest X-ray images dataset.
- Identification of majority-class collapse and clinical safety failures of VGG-19 and Swin-Tiny under realistic class imbalance.
- Stratified 5-fold cross-validation with McNemar's pairwise significance testing across all 45 model pairs.
- Comprehensive measurements and hardware-aware deployment analysis including latency, throughput, VRAM consumption and FLOPs.
- Explainability comparison using Grad-CAM++ between robust and collapsed architectures.

II. LITERATURE REVIEW

The deep learning models have a larger impact on medical image analysis and evaluations. The research studies have revealed that there are hundreds of applications of image analysis in the medical domain that have been developed using deep learning convolutional neural networks, including image detection, segmentation, and classification. This

remarkable usage and contribution of deep learning models have played a very positive role in the image analysis and diagnosis problems in the medical field; because of the use of computer-aided diagnosis with efficiency and accuracy in decision making, many researchers have claimed that the CAD has mitigated the shortage of radiologist staff and technical resources in laboratories and hospitals [13].

A. MACHINE LEARNING MODELS

Before the deep learning era, pneumonia detection relied on handcrafted features. Traditional machine learning approaches such as Random Forests (RFs), Artificial Neural Networks (ANNs), Support Vector Machines (SVMs), and k-Nearest Neighbors (k-NN) extracted statistical features from segmented lung [14], [15]. These approaches suffered from limited generalizability due to small dataset size [16]; and the inherent constraints of hand-engineered feature pipelines. The primarily reason of less performance of these machine learning classifiers is that they were more dependent on the quality of manually extracted features [17]. As a result, most common machine learning techniques encountered severe challenges in obtaining the reliable accuracy and consistent performance for pneumonia detection, which led to the shift toward the deep learning-based models which automatically learn from complex data and discriminative features directly from medical images [18].

B. CONVOLUTIONAL NEURAL NETWORKS (CNNS) MODELS

The introduction of CheXNet research study with use of DenseNet-121 architecture ChestX-ray 14 image dataset established breakthrough in automated chest radiograph interpretation. This research evaluated various diseases, including thoracic and pneumonia in the medical field [19]. This research used a large-scale chest X-ray-14 dataset and achieved a more robust performance like a radiologist, for pneumonia detection and witnessed the higher performance from an average radiologist in the pneumonia detection from the X-ray images dataset. The research in [20] revealed that CNNs are very effective for the detection of tuberculosis disease from chest x-ray images and demonstrated their performance on other diseases, including pneumonia, and concluded the architectures generalization. Two other researchers, Liang and Zheng, claimed improved performance and accuracy in the pneumonia detection task through their proposed CNN-based framework using a chest X-ray images dataset [21]. Liang and Zheng [22] further introduced dilated convolution to preserve spatial information while controlling overfitting. Furthermore, a deep learning-based system proposed by Kermany et al in [23] for pediatric pneumonia diagnosis obtained higher accuracy and described the potential of CNN architectures in clinical practices and in real-world applications in the medical domain. In another study, conducted by Hammoudi et al. [24] examined deep learning methods for analyzing and evaluating chest X-ray images to detect pneumonia disease, specifically in the context of COVID-19 detection and screening. While, Gabruseva et al. [25] proposed and developed a deep learning method for pneumonia region detection based on single-shot detectors with SE-ResNeXt101 encoders, and claimed to achieve competitive outcomes on the RSNA Pneumonia Detection Challenge. These studies of convolutional neural networks contributed extensively into medical diagnostic systems using deep learning.

C. TRANSFER LEARNING

The emergence of transfer learning has contributed significantly to deep learning, specifically for pneumonia detection from chest X-ray images. Transfer learning is useful when there is limited availability of labelled datasets for pneumonia detection and classification. Transfer learning allows Pre-trained models like ImageNet to be fine-tuned to perform the desired task for classification and detection on a limited dataset of pneumonia disease. In [26] critically experimented and evaluated the efficiency of transfer learning from natural images to medical imaging and concluded that transfer learning is widely used in many applications; its benefits may vary depending on the task complexity. Another research conducted by Rahman et al. [27] systematically experimented and evaluated transfer learning with AlexNet, ResNet-18, DenseNet-201, and SqueezeNet. Similarly, Bassi and Attux [28] applied DenseNet architectures with twice transfer learning and output neuron keeping, achieving a test accuracy of 100% on their COVID-19 detection dataset. The further analysis of transfer learning strategies in medical image analysis was conducted, highlighting its role in enhancing and improving diagnostic performance under a limited image dataset [29]. The transfer learning continuously improves stability, convergence speed, and classification performance.

D. VISION TRANSFORMERS

Recent research studies suggest that vision transformers (ViTs) have emerged as a useful alternative to convolutional neural networks (CNNs) by using self-attention mechanisms to grab long-range dependencies and global contextual information within images. Dosovitskiy et al. [11] evaluated that transformers are more powerful to obtain effective results on image classification tasks in medical imaging, when applied directly to sequences of image patches on a large-scale image dataset. Though vision transformers face some challenges of limited locality and high dimensionality, they are more dependent on large image datasets. Furthermore, the Attention U-Net introduced attention gates to focus on relevant regions of medical images, improving feature localization and interpretability [30]. Moreover, one more research conducted [31] highlighted that the hybrid CNN-Transformer models combine convolutional feature extraction with transformer-based global reasoning, achieving state-of-the-art performance for pneumonia detection and classification from chest X-ray images. Furthermore, in order to overcome the problem of more dependency of ViT on large-scale images dataset, one of the researchers, Touvron et al. [32] suggested the Data-efficient Image Transformer

(DeiT), which uses wide-ranging regularization, data augmentation, and knowledge extraction approaches to empower effective transformer training and learning on limited images datasets. This significant development in vision transformers research reduced their dependency on large-scale pre-training and enhanced the experimental results of Vision Transformers for different computer vision tasks. Many research studies consequently emphasized improving local feature extraction and computational efficiency. Similarly, Li et al., [33] presented the Tokens-to-Token Vision Transformer (T2T-ViT), which progressively aggregates neighboring tokens to save local structural information and make more informative visual representations. Likewise, another research conducted by Han et al. [34] proposed Transformer-in-Transformer (TNT), a hierarchical architecture that jointly simulates the pixel-level and patch-level features, allowing improved capture of both local and global characteristics from the images. Additionally, in order to report the requirement for multi-scale feature illustration, Wang et al., [35] experimented and proposed the Pyramid Vision Transformer (PVT), which utilizes a hierarchical pyramid structure type to generate feature maps from images at multiple resolutions. Using the structure of this framework, Wang et al., [36] established a PVTv2 by joining overlapping patch embedding, linear-complexity attention mechanisms, and convolution-enhanced feed-forward networks, producing an enhanced feature extraction mechanism and computational efficiency. Liu et al., [12] introduced a key advancement in hierarchical transformer design via the Swin Transformer. This research applied the techniques of hierarchical feature maps and shifted-window self-attention and claimed that the Swin Transformer achieves linear computational complexity with respect to image size, though maintaining robust performance in multiple tasks, including segmentation, classification, and detection from images. Other than architectural improvements and developments, Raghu et al. [37] probed and demonstrated the representational characteristics of Vision Transformers and established that the transformer-based models largely learn different feature representations from images as compared to CNNs, with improved protection of global contextual information and improved transfer learning capabilities. Moreover, multiple research studies have demonstrated that Vision Transformers exhibit solid performance across many medical imaging tasks, including lesion detection, disease classification, and diagnostic decision support, highlighting their emerging importance in many healthcare systems [38], [39]. Moreover, Radočaj et al., [40] introduced adaptive multi-scale feature learning modules that improved recognition of subtle pulmonary infiltrates by capturing lesion information at multiple receptive fields. Salamon and Ksiażek [41] investigated capsule neural networks optimized through Bayesian search and reported improved sensitivity for pediatric pneumonia detection. Comparative studies have further reinforced the value of transfer learning. Benmalek [42] demonstrated that Xception outperformed traditional VGG and ResNet variants on pediatric chest radiographs, while Choudhury [43] showed that fine-tuned ResNet50 significantly outperformed custom CNN architectures. Similarly, Hosseinabadi and Rohani [44] reported superior performance of RegNet architectures compared with EfficientNetV2 and ResNetRS. Hybrid CNN-Transformer approaches have also gained momentum. LungX [45] combined EfficientNet feature extraction with Vision Transformer attention mechanisms and achieved improved lesion localization and classification performance. Xu and Zhang [46] addressed class imbalance through GAN-based enhancement techniques; whereas, Barman and Giri [47] proposed localized patch ensembles that improved diagnostic accuracy through region-specific feature extraction. Recently, Sikindar et al., [48] demonstrated that multimodal fusion strategies based on Swin Transformers can further enhance pneumonia detection by integrating complementary imaging representations. Generally, these advancements reveal the rapid growth of Vision Transformer architectures toward improved multi-scale feature extraction, locality modeling, computational efficiency, and medical imaging performance. Nevertheless, gaps remain regarding systematic CNN-versus-ViT benchmarking, statistical significance testing, failure-mode analysis, explainability validation, and computational efficiency profiling. The proposed framework addresses these shortcomings through large-scale evaluation, McNemar statistical testing, clinical safety analysis, explainability via Grad-CAM++, and comprehensive hardware-aware benchmarking. This study addresses all these highlighted gaps through a rigorous experimental design and evaluation.

E. EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI)

Commonly deep learning models behaves like black box predictors, so to interpret their outcome visually, we utilize the Explainability which has become very important and critical for trustworthy AI medical diagnostic systems. As clinicians may avoid adopting black box value predictions to automate their decision-making systems without solid reasoning. To overcome and address this challenge, Explainable Artificial Intelligence (XAI) methods have been widely used and tested to visualize medical image regions in final predictions. The common approaches used to measure the deep learning based diagnostic systems are Gradient-weighted Class Activation Mapping (Grad-CAM) and Grad-CAM++ [49], [50]. These approaches create visual representation in the form of heatmaps which mark the clinically related regions in the image which ensures the verification that model decisions are based on pathological findings rather than unauthentic image artifacts.

F. CLINICAL SAFETY

The increasing need and deployment of artificial intelligence applications in healthcare has emphasized on clinical safety assessment and evaluation. While several research studies have been conducted on classification accuracy for pneumonia detection. However, only accuracy does not assure the clinical safety, specifically, when there is imbalanced dataset.

Deep learning models may demonstrate different unseen failure modes, poor generalization through patient population, majority-class prediction collapse, and unsteady performance under clinically genuine class distributions [51]. Such proven catastrophes can lead to very high false-negative outcomes, possibly deferring treatment and diagnosis of pneumonia. Therefore, Deep learning model's assessment and evaluation in medical applications should extend further and consider statistical reliability, explainability, failure mode analysis, and deployment proficiency not only prediction accuracy. These considerations are dire need in pneumonia diagnostic applications, where lack of patient diagnosis may result to delayed treatment and adversarial outcomes.

Therefore, this proposed research highlights these gaps through a large-scale experimental benchmarking framework that incorporates diagnostic performance evaluation, McNemar statistical testing, computational efficiency profiling, Grad-CAM++ explainability validation, and clinical safety analysis using a combined pediatric-adult chest X-ray dataset.

III. RESEARCH METHODOLOGY

This research is based on experimental design with an overall architecture consisting of a multi-stage pipeline from chest X-ray input to clinical output. Starting from data acquisition and preprocessing datasets, 10-model parallel training with stratified 5-fold cross-validation, Performance evaluation, computational efficiency profiling, and explainability via Grad-CAM++ as shown in Figure 1.

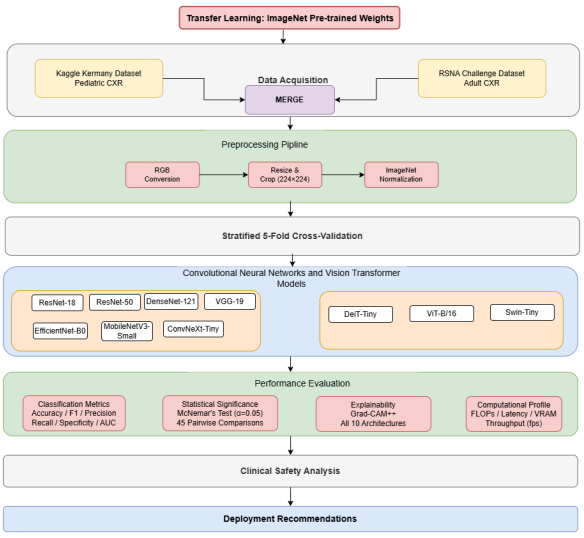


FIGURE 1: System Architecture from chest X-ray image acquisition to clinical decision support output.

A. DATA ACQUISITION

We combined two publicly available chest X-ray image datasets to maximize sample diversity and clinical generalizability. The first dataset, which contains pediatric chest X-ray images of size 5856 samples, was introduced by Kermany [23], and the second dataset of adult chest X-ray images was obtained from the RSNA Pneumonia Detection Challenge dataset of size 25,684 images and originally derived from the NIH chestX-ray dataset [53]. The details of both datasets are mentioned in Table 1.

TABLE 1: Mixed Clinical Chest X-ray dataset composition

Source	Total	Normal	Pneumonia	Type
Kaggle Kermany	5,856	1,583	4,273	Pediatric
RSNA Challenge	25,684	20,025	5,659	Adult
Combined	**31,540**	**21,608**	**9,932**	**Mixed**

This is a combined dataset which contains a 68.5% Normal 31.5% Pneumonia, which is a slight imbalance; its class Imbalance Ratio (IR) can be calculated as:

$$IR = \frac{N_{majority}}{N_{minority}} \tag{1}$$

This moderate imbalance ($IR = 9,932/21,608 = 2.18$), genuinely reflects clinical exercise where most hospital chest X-ray images CXRs are normal and do not require synthetic balancing for modern architectures; and it also discloses weak architectures which has been evaluated in this research including VGG-19 from CNNs and Swin-Tiny from Transformers family are collapsing to majority class prediction, which is one of the clinically unsafe challenges has been highlighted in this research study.

B. TRANSFER LEARNING STRATEGY

Training deep neural networks from scratch for medical image analysis often requires extremely large annotated datasets and substantial computational resources. To improve convergence, reduce overfitting, and leverage previously learned visual representations, transfer learning was employed for all evaluated architectures. Specifically, all Convolutional Neural Networks (CNNs) and Vision Transformer (ViT) models were initialized using weights pretrained on the ImageNet dataset, which contains more than one million natural images across 1,000 object categories. The transfer learning process can be expressed as:

$$W_{fine} = W_{imgNet} + \Delta W \tag{2}$$

Where, W_{imgNet} represent the ImageNet pre-trained weights, ΔW shows the learned updated during pneumonia training, and W_{fine} represents the final optimized weights. Unlike feature-extraction approaches that freeze a portion of the network, this study adopted an end-to-end fine-tuning strategy. All layers of each architecture were initialized using pretrained ImageNet weights and subsequently updated during training. Consequently, every trainable parameter contributed to adapting generic visual features to domain-specific radiological characteristics associated with pneumonia detection. The optimization can be represented mathematically as:

$$W^* = \arg\min_{W} L(W) \tag{3}$$

Where W shows the set of all trainable model parameters (weights and biases), $L(W)$ expresses the loss function evaluated using parameter set W, and W^* represents the optimal parameter set that minimizes the loss function and produces the best model performance on the training data.

C. IMAGE PREPROCESSING PIPELINE

To ensure effective fine-tuning of deep learning models originally pre-trained on the ImageNet dataset, a consistent image preprocessing mechanism was adopted in this research. The three major phases of this preprocessing pipeline are image Red Green Blue (RGB) conversion, image resizing, and image normalization, along with their clinical justifications, and visual illustrations are mentioned the below sections.

1) Image RGB Conversion

Common deep learning architectures, including ResNet, EfficientNet, and Vision Transformers, are built to accept three-channel RGB image inputs. However, the images dataset 2 which was collected from RSNA Pneumonia Detection Challenge, is in a single channel gray scale images format. To meet with the architectural requirements of these models and for training purposes the single channel images are converted to three channel representation through bridge this architectural gap, single-channel inputs are converted to three-channel representations by duplicating the grayscale intensity values across all three-color channels (R = G = B). Replicating the single channel does not mean to add new color information or change the diagnostic content of the image. However, it is mathematically required to use ImageNet-pretrained weights, allowing these deep learning models to pull general image features like edges, textures, and shapes learned from natural image databases. The mathematical representation of RGB conversion is mentioned in equation (4).

$$I_{RGB}(x, y) = [I(x, y), I(x, y), I(x, y)] \tag{4}$$

The image RGB conversion samples are mentioned in the Figure 2 below.

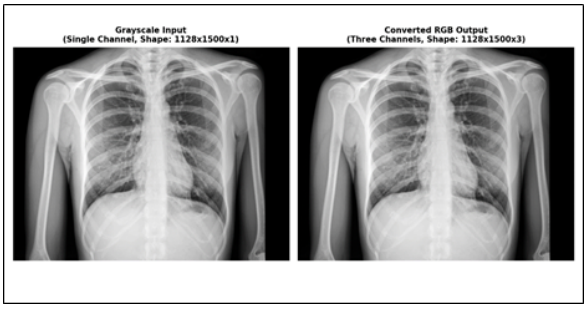


FIGURE 2: Grayscale input representation mapped to 3-channel RGB dimensions

2) Image Resizing and Cropping

Chest X-rays in raw formats can have resolutions as high as 1024×1024 pixels. Direct training on these high resolutions imposes prohibitive computational costs and memory overhead. To mitigate this, images are resized to a standard bounding box. The image resizing is represented as:

$$Ir = Resize(I, H, W) \tag{5}$$

During training, a random resized crop is used for spatial data augmentation. During validation and testing, center cropping is used to obtain a clean, deterministic focal area containing the lung fields. Resizing to a uniform size ensures the model processes images with matching spatial hierarchies. Center cropping removes peripheral areas containing scan markers, labels, or scanner borders that could introduce bias, while keeping the clinical focus centered on the lung parenchyma.

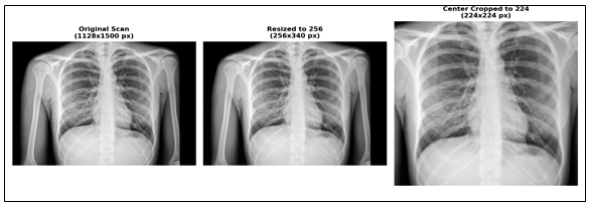


FIGURE 3: Resizing and cropping operations mapping input scans to standard 224×224 format.

3) Image Normalization

The raw pixel intensity values of RGB images typically range from 0 to 255. Before training, image normalization is applied to standardize the pixel distributions across channels. First, pixel values are scaled to the range $[0, 1]$, after, which channel-wise normalization is performed through:

$$x_{norm} = \frac{x - \mu}{\sigma} \tag{6}$$

where x denotes the input pixel value, μ represents the channel mean, and σ denotes the channel standard deviation. To ensure compatibility with the pretrained network weights, normalization is performed using the ImageNet dataset statistics: $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$ for the red (R), green (G), and blue (B) channels of images, respectively. Applying ImageNet-specific normalization shifts the images intensity distributions to match those the model expects as shown in the Figure 4 below. This prevents gradient explosion or vanishing, stabilizes the optimization process, and accelerates loss convergence during transfer learning.

Standard train/validation/test splits can produce high variance in performance estimates due to random partitioning.

D. CROSS VALIDATION

Stratified k-fold cross-validation which preserves class distribution in each fold, provides more robust performance

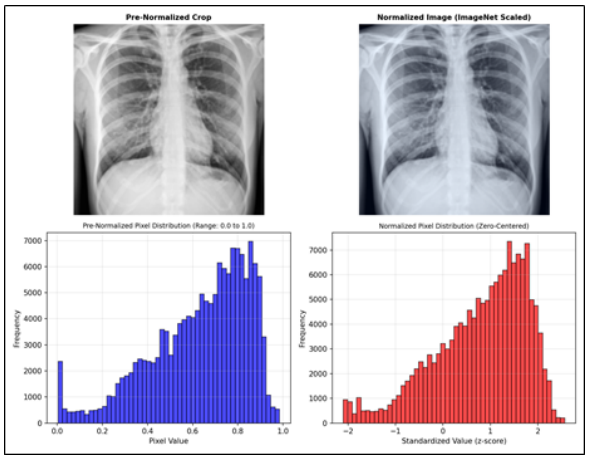


FIGURE 4: Pre-normalized vs. normalized image results and their corresponding pixel distribution shifts.

estimates, and enables statistical comparison using paired significance tests. The entire images dataset was pooled and stratified by class label. Five folds were created with 25,232 training images (80%) and 6,308 validation images (20%) per fold. Each fold trained a fresh model from ImageNet-pretrained weights, ensuring no data leakage between folds. The cross validation performance metric was calculated as:

$$CV_Score = \frac{1}{K} \times \sum_{i=1}^{K} Score(M_i, D_{val}^i) \tag{7}$$

The classification accuracy of each fold was computed as, where, TP, TN, FP, and FN represent the true positive, true negative, false positive, and false negative for fold which is represented i, respectively.

$$Acc_i = \frac{TP_i + TN_i}{TP_i + TN_i + FP_i + FN_i} \tag{8}$$

Whereas, the mean performance for each fold across for all models is calculated in equation (9), and it is represented with μ, and standard deviation was calculated with equation (10).

$$\mu = \frac{1}{K} \times \sum_{i=1}^{K} M_i \tag{9}$$

$$\sigma = \sqrt{\frac{1}{(K-1)} \times \sum_{i=1}^{K} (M_i - \mu)^2} \tag{10}$$

which is represented with σ, where, M_i represents the performance metric (accuracy, F1-score, precision, recall, specificity, or AUC) for fold i. After that the F1-score which plays important role in balancing the precision and recall values, and was computed for each fold is represented as:

$$F1_i = 2 \times \frac{Precision_i \times Recall_i}{Precision_i + Recall_i} \tag{11}$$

where,

$$Precision_i = \frac{TP_i}{TP_i + FP_i} \tag{12}$$

$$Recall_i = \frac{TP_i}{TP_i + FN_i} \tag{13}$$

$$AUC = \int_0^1 TPR(t) \cdot d(FPR(t)) \tag{14}$$

Where, the equation (14) of AUC $TPR(t)$ represents the true positive rate and $FPR(t)$ shows the false positive rate at threshold (t) respectively.

IV. NEURAL NETWORKS AND VISION TRANSFORMERS

A. CONVOLUTIONAL NEURAL NETWORKS (CNN) ARCHITECTURE

This research paper presents a benchmark framework integrating 10 architectures, seven Convolutional Neural Networks (CNNs) as mentioned in the Table 2, including ResNet-18/50, DenseNet-121, VGG-19, EfficientNet-B0, MobileNetV3-Small, ConvNeXt-Tiny.

TABLE 2: CNN architectures evaluated in this study

Model (Year)	Key Innovation	Params
ResNet-18 (2015)	Residual connections, BatchNorm	11.18M
ResNet-50 (2015)	Deeper residual bottleneck blocks	23.51M
DenseNet-121 (2017)	Dense connectivity, feature reuse	6.96M
VGG-19 (2014)	Deep sequential conv. (no BN)	139.58M
EfficientNet-B0 (2019)	Compound scaling (W×D×R)	4.01M
MobileNetV3-Small (2019)	Inverted residuals, SE	1.52M
ConvNeXt-Tiny (2022)	Modernized ResNet with Transformer	27.82M

The ResNet with skip connections to address the vanishing gradient problem, enabling training of networks with up to 152 layers. The residual learning is represented as:

$$y = F(x, \{W_i\}) + x \tag{15}$$

Where $F(x, \{W_i\})$ represents the residual mapping, and x works like the input to the skip-connection. This lacks batch normalization and residual connections, making it susceptible to optimization difficulties on imbalanced datasets [52].

B. VISION TRANSFORMER ARCHITECTURES

This research compares the CNNs models with Vision Transformers through experiments via three Vision Transformers (DeiT-Tiny, Swin-Tiny, ViT-B/16) using a 31,540 Chest X-ray (CXRs) image dataset as mentioned in the Table 3 below.

TABLE 3: Vision Transformer architectures evaluated

Model (Year)	Key Innovation	Params
DeiT-Tiny (2021)	Data-efficient ViT with distillation	5.52M
Swin-Tiny (2021)	Shifted window self-attention	27.52M
ViT-B/16 (2020)	Pure ViT, 16×16 patches	85.80M

ViT-B/16 (85.80M parameters) represents the standard Vision Transformer configuration: 12 transformer layers, 768-dimensional embedding, 12 attention heads, and 16×16 pixel patches. Each image (224×224) is divided into 196 non-overlapping patches, linearly projected to token embeddings, augmented with a learnable [CLS] token and position embeddings, and processed through L stacked Transformer encoder layers. The Swin Transformer addresses ViT's computational limitation through local window attention with cross-window connections, achieving linear rather than quadratic complexity with respect to image size. The multi-head self-attention mechanism demonstrated in various research studies and central to all ViT variants is formulated as:

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{16}$$

$$MultiHead(Q, K, V) = Concat(head_1, ..., head_h)W^O \tag{17}$$

where Q, K, V are query, key, and value matrices, d_k is the dimension of keys, and h is the number of attention heads. This global attention mechanism is a key differentiator from CNN locality assumptions.

C. TRAINING CONFIGURATION

All models used identical hyperparameters to ensure fair comparison and reproducibility as shown in Table 4:

TABLE 4: Unified training configuration

Parameter	Value
Optimizer	Adam (β_1=0.9, β_2=0.999)
Init. Learning Rate	0.001
Weight Decay	1×10^{-4}
LR Scheduler	ReduceLROnPlateau (factor=0.5, patience=3)
Loss Function	CrossEntropyLoss
Mixed Precision	FP16 with GradScaler
Gradient Clipping	max_norm = 1.0
Max Epochs	50
Early Stopping	Patience = 8 epochs
Batch Size	64

D. EVALUATION METRICS

All proposed models were evaluated using standard classification metrics appropriate for binary classification. These evaluation metrics offer a complete assessment of the model's capability to correctly differentiate between positive and negative classes from pneumonia images. The primary metrics on which our 10 models have been tested, including Accuracy, Precision, Recall, Specificity, f1-score, Area Under Curve (AUC) has been computed and represented mathematically in following equations.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{18}$$

$$Precision = \frac{TP}{TP + FP} \tag{19}$$

$$Recall = \frac{TP}{TP + FN} \tag{20}$$

$$Specificity = \frac{TN}{TN + FP} \tag{21}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{22}$$

Where True Positives are denoted by (TP), true negatives are represented with (TN), false positives are denoted with (FP), and (FN) means false negatives and True Positive Ratio (TPR) denotes the False Positive Rate.

E. MCNEMAR'S STATISTICAL SIGNIFICANCE TESTING

To evaluate and compare the performance of classifiers, the McNemar's statistical significance test is conducted on the same test set. The results states that both tested classifiers have the equal error value. Mathematically represented as:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \tag{23}$$

In this study McNemar's test has been applied pairwise on all 45 model combinations with value ($\alpha = 0.05$) which is statistical significance.

F. COMPUTATIONAL EFFICIENCY METRICS

This research has also measured the efficiency metrics to validate the clinical deployment and decisions regarding these models which is mentioned in the Table 5.

TABLE 5: Computational efficiency metrics

Metric	Definition	Tool
FLOPs	Floating point ops per forward pass	thop
Latency (ms)	Time per single image	PyTorch
Throughput (fps)	Images processed per second	1000/Latency
VRAM (MB)	GPU memory consumption	nvidia-smi

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. MODELS TRAINING RESULTS ANALYSIS

The experimental results including the training and validation outcomes for folds are illustrated in this section, which show overfitting, convergence behavior, and training performance of all these deep learning architectures. The following figures show the training and validation accuracy and loss curves for representative folds (Fold 1) of each architecture during 5-fold cross-validation.

Training and validation accuracy and loss curves for representative folds. Six dual-panel figures (one per representative architecture) in the above given figures (Left) Training and validation accuracy over epochs on (Right) Training and validation loss over epochs. ResNet-18 (Figure 5), DenseNet-121 (Figure 6), and EfficientNet-B0 (Figure 8) show stable convergence with tight train-validation curves. VGG-19 (Figure 7) and Swin-Tiny (Figure 9) exhibit abrupt validation

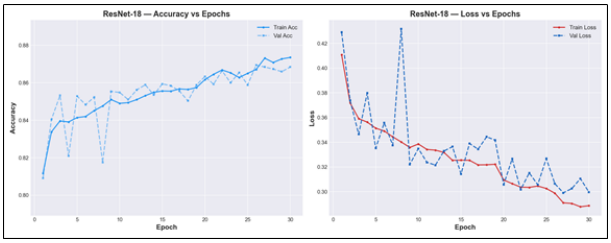


FIGURE 5: ResNet-18 Training and validation accuracy and loss over epochs

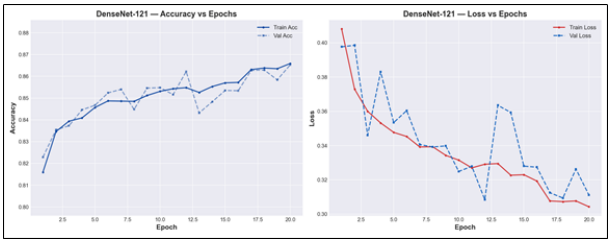


FIGURE 6: DenseNet-121 Training and validation accuracy and loss over epochs

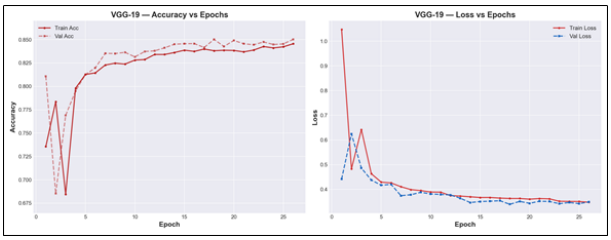


FIGURE 7: VGG-19 Training and validation accuracy and loss over epochs

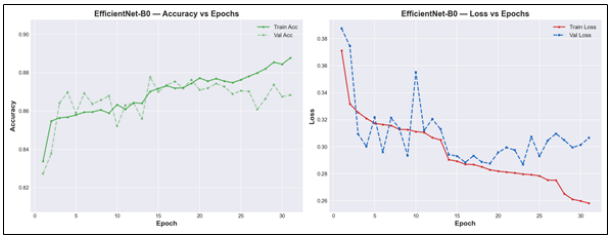


FIGURE 8: EfficientNet-B0 Training and validation accuracy and loss over epochs

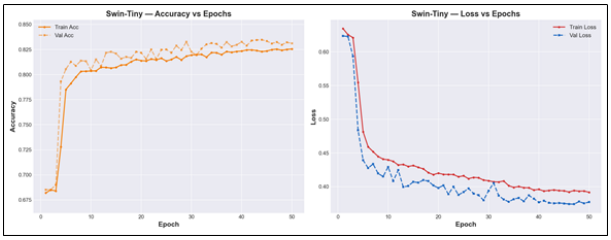


FIGURE 9: Swin-Tiny Training and validation accuracy and loss over epochs

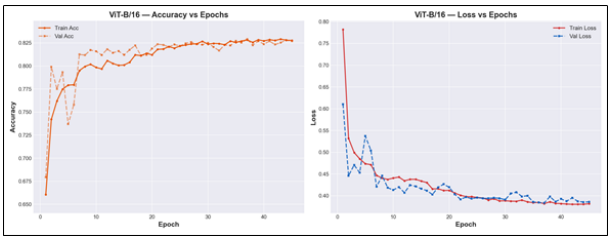


FIGURE 10: ViT-B/16 Training and validation accuracy and loss over epochs

metric collapse. ViT-B/16 (Figure 10) shows slower convergence with higher validation variance. It is very important to note that MobileNetV3-Small, ConvNeXt-Tiny, ResNet-50, and DeiT-Tiny follow similar convergence patterns to their respective architecture families.

B. FIVE-FOLD CROSS-VALIDATION EXPERIMENTAL RESULTS

Table 6 presents the comprehensive performance by comparing all 10 architectures across stratified 5-fold cross-validation. Table 6 shows that Swin-Tiny and VGG-19 exhibited majority-class collapse in 3/5 folds; large SDs indicate fold instability, and the same is highlighted with * in Table 6 below. The values are reported as mean ± standard deviation across 5 folds.

EfficientNet-B0 achieved the highest mean accuracy (86.9%) with the smallest variance (±0.004) among the top-5 models. The progressive performance gap between CNNs and ViTs is clearly visible from the experimental results mentioned in Table 6 above, which indicates that the best ViT (DeiT-Tiny) achieves only 83.7% accuracy, 3.2 percentage points below EfficientNet-B0 ($p < 0.001$, McNemar's test). The extreme standard deviations for VGG-19 (±0.089) and Swin-Tiny (±0.076) reflect disastrous majority-class collapse across multiple folds.

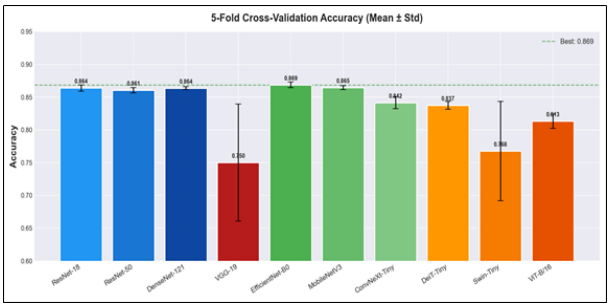


FIGURE 11: 5-Fold Cross-Validation Accuracy (Mean ± SD) for all 10 architectures. Error bars represent standard deviation across folds.

Figure 11 demonstrates the 5-Fold Cross-Validation Accuracy for all 10 architectures. Bar chart with error bars (±1 SD). EfficientNet-B0 achieves the highest mean accuracy (86.9%) with the smallest variance (±0.004). VGG-19 and Swin-Tiny exhibit large error bars, indicating fold instability.

TABLE 6: Performance comparison of 10 architectures (mean ± SD, 5-fold CV)

Rank	Model	Accuracy	F1 Score	Precision	Recall	Specificity	AUC
1	EfficientNet-B0	0.869±0.004	0.777±0.010	0.835±0.004	0.726±0.018	0.934±0.003	0.932±0.006
2	MobileNetV3-Small	0.865±0.003	0.766±0.010	0.841±0.020	0.705±0.029	0.938±0.012	0.929±0.003
3	ResNet-18	0.864±0.005	0.767±0.010	0.831±0.013	0.713±0.018	0.933±0.007	0.927±0.006
4	DenseNet-121	0.864±0.002	0.763±0.004	0.842±0.005	0.699±0.005	0.940±0.002	0.926±0.005
5	ResNet-50	0.861±0.004	0.761±0.011	0.826±0.009	0.707±0.022	0.931±0.006	0.921±0.009
6	ConvNeXt-Tiny	0.842±0.009	0.726±0.022	0.798±0.034	0.668±0.047	0.921±0.018	0.904±0.011
7	DeiT-Tiny	0.837±0.006	0.720±0.009	0.788±0.023	0.663±0.020	0.917±0.013	0.897±0.006
8	ViT-B/16	0.813±0.011	0.675±0.019	0.748±0.031	0.617±0.033	0.903±0.020	0.864±0.013
9	Swin-Tiny *	0.768±0.076	0.408±0.373	0.473±0.431	0.360±0.331	0.955±0.042	0.723±0.203
10	VGG-19 *	0.750±0.089	0.296±0.406	0.320±0.438	0.276±0.378	0.968±0.043	0.662±0.223

Statistical significance testing (McNemar's test, $\alpha = 0.05$) showed EfficientNet-B0 significantly outperforms all other models ($p \leq 0.004$).

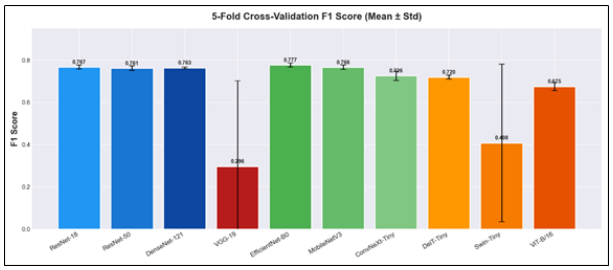


FIGURE 12: F1 Score comparison across all 10 architectures

Figure 12 indicates that the F1 Score Comparison across all 10 architectures. Bar chart showing mean F1 scores. VGG-19 and Swin-Tiny show near-zero F1 (0.296 and 0.408, respectively) due to majority-class collapse. EfficientNet-B0 leads with 0.777 F1, followed by ResNet-18 (0.767) and MobileNetV3-Small (0.766).

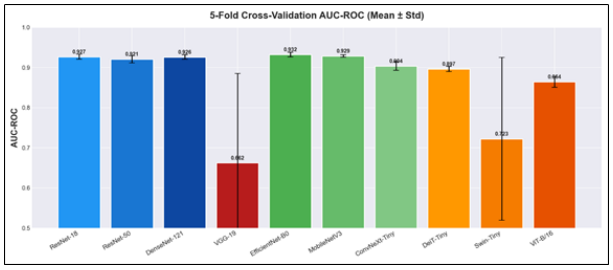


FIGURE 13: AUC-ROC comparison across all 10 architectures.

AUC-ROC Comparison Across all 10 architectures as mentioned in the Figure 13. Bar chart showing mean AUC values. EfficientNet-B0 leads with 0.932 AUC, followed by MobileNetV3-Small (0.929) and ResNet-18 (0.927). AUC near 0.5 for VGG-19 (0.662) and Swin-Tiny (0.723) confirms near-random classification due to collapse. In general, AUC values near 0.5 for collapsed models indicate failure equivalent to random classification.

Figure 14 shows a precision vs. Recall Scatter Plot. Precision (x-axis) vs. Recall (y-axis) for all 10 models. Top-right cluster (high precision, high recall): EfficientNet-B0, MobileNetV3-Small, ResNet-18, DenseNet-121, ResNet-50. Bottom-left isolated cluster (low precision, low recall): VGG-19 and Swin-Tiny confirming majority-class collapse. ConvNeXt-Tiny, DeiT-Tiny, and ViT-B/16 occupy intermediate positions. The spatial separation clearly distinguishes stable CNNs from collapsed models.

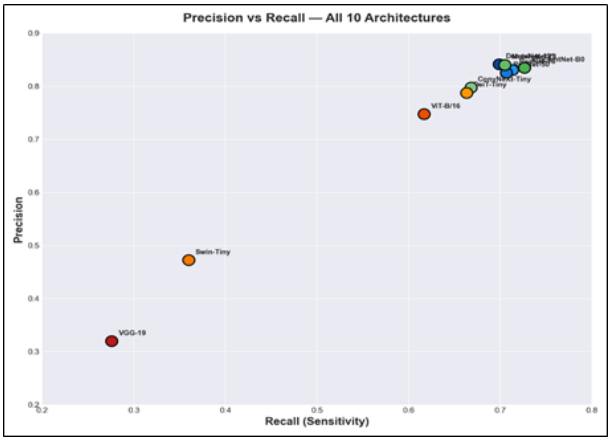


FIGURE 14: Precision–Recall scatter plot for all 10 architectures

C. COMPUTATIONAL EFFICIENCY PROFILING

Figure 15 shows model size (Parameters) vs. Accuracy. Scatter plot with $\log_{10}$(parameters) on the x-axis and accuracy on the y-axis. EfficientNet-B0 achieves the best accuracy with 4.01M parameters.

VGG-19 has 139.58M parameters (28× larger) with lower accuracy and collapse risk.

As illustrated in the Figure 16 which depicts the comparison between computational cost and Accuracy results. Lightweight models (EfficientNet-B0, 0.83 GFLOPs; MobileNetV3-Small, 0.12 GFLOPs; ResNet-18, 3.65 GFLOPs) dominate the upper-left (efficient + accurate) region. VGG-19 and ViT-B/16 both models require higher computational costs (GFLOPs) without accuracy gain. On the other hand lightweight convolutional neural networks (CNNs) hold balance performance between computational cost vs accuracy on upper-left region. In conclusion dense deep learning models are not appropriate with respect to accuracy benefit.

TABLE 7: Computational efficiency profiling on NVIDIA RTX 5060 Ti

Model	Params (M)	GFLOPs	Latency (ms)	Throughput (fps)	VRAM (MB)
MobileNetV3-Small	1.52	0.12	8.28	120.8	45.3
EfficientNet-B0	4.01	0.83	12.50	80.0	63.0
DeiT-Tiny	5.52	2.15	8.43	118.6	60.4
DenseNet-121	6.96	5.79	22.67	44.1	47.1
ResNet-18	11.18	3.65	1.75	572.7	67.8
ResNet-50	23.51	8.26	3.96	252.4	161.6
Swin-Tiny	27.52	8.74	18.32	54.6	161.4
ConvNeXt-Tiny	27.82	8.93	8.63	115.9	163.9
ViT-B/16	85.80	33.70	6.42	155.9	478.8
VGG-19	139.58	39.26	4.34	230.6	619.1

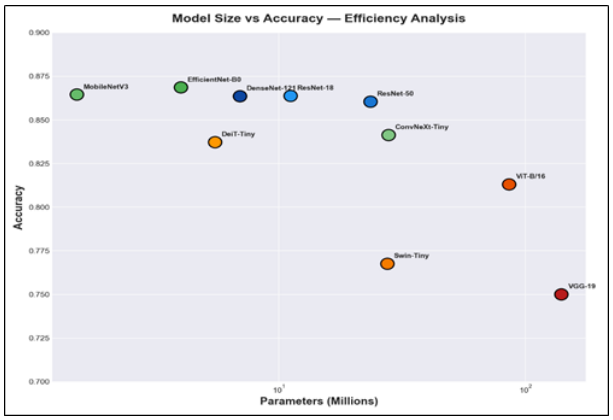


FIGURE 15: Parameter count (log scale) vs. Accuracy

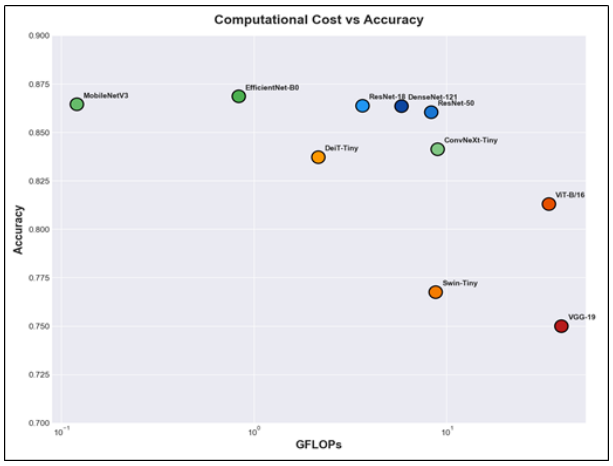


FIGURE 16: Illustration of computational cost (GFLOPs) vs Accuracy Measurement

The Figure 17 shows scatter plot between Inference Speed (fps) and Accuracy measurement. The experimental results in the graph show that the ResNet-18 achieves 572.7 fps with 86.4% accuracy and good choice for real-time deployment. Moreover, the EfficientNet-B0 achieved highest 86.9% accuracy on 80 fps. On other hand, the MobileNetV3-Small obtains 120.8 fps with values of 86.5% accuracy, which is again a reasonable option for mobile devices deployment. In conclusion from results that the ResNet-18 proposes the ideal speed vs accuracy tradeoff for real-time clinical applications

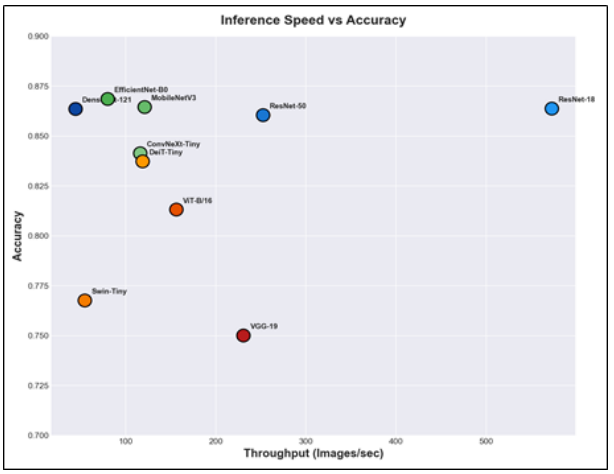


FIGURE 17: Performance Illustration of Inference throughput (fps) vs Accuracy result

in healthcare systems.

D. MCNEMAR'S PAIRWISE STATISTICAL SIGNIFICANCE

We have performed the depth experimental McNemar's pair wise testing for all 45 models and focused on the comparison analysis of statistical significance difference between them. In result it is concluded that only three model pairs resulted statistically non-significant difference ($p > 0.05$) as shown in the Table 8.

TABLE 8: Result analysis of non-significant deep learning model pairs through McNemar's pairwise testing

Model A	Model B	p-value	Significant?
ResNet-18	DenseNet-121	0.889	No
ResNet-18	MobileNetV3-Small	0.646	No
DenseNet-121	MobileNetV3-Small	0.536	No

The testing result mentioned in the Table 8 also shows that all other 42 model pairs were statistically significant ($p < 0.05$). The statistically significant ($p \leq 0.004$) shows that the EfficientNet-B0 was better than all other deep learning models which shows that this model has performed remarkably well among all other trained models.

Figure 18 illustrates the six-evaluation metrics including Accuracy, F1, Precision, Recall, Specificity, and AUC of top-five deep learning architectures and their performance

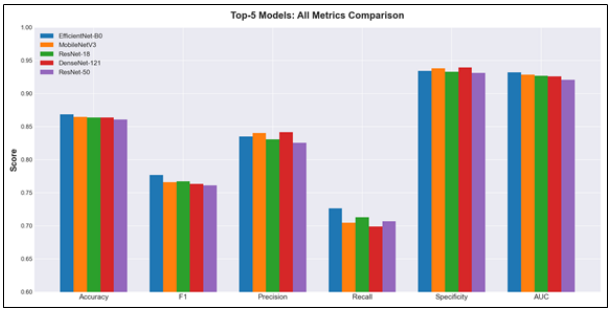


FIGURE 18: The experimental Results of evaluation metrics of Top-5 Architectures

comparison through a multi-metric radar chart. It is evident from the experimental results that the EfficientNet-B0 maintained highest or second-highest balanced consistently and superior in performance in all metrics simultaneously. Similarly, the MobileNetV3-Small also achieves good balanced performance throughout all evaluation metrics.

E. EXPLAINABILITY (GRAD-CAM++ VALIDATION) TESTING

The explainability experimental and visualization results through Grad-CAM++ approach ensures models success in clinical decision support. It also confirms that EfficientNet-B0, ResNet-18, and DenseNet-121 architectures particularly focus on lung parenchyma regions and proving their reliability with radiological practices in healthcare. The important observations were highlighted through explainability that:

- For normal X-ray images, attention is drawn-out diagonally for the lung fields with no central concentration as shown in the Figure 19.
- For pneumonia X-ray images, model attention concentrates on particular region in the x-ray image, and correctly identifying pneumonia affected denser areas from the x-ray image as shown in the Figure 20.

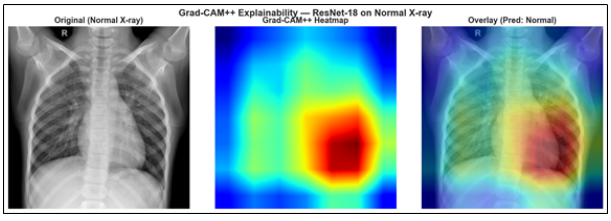


FIGURE 19: The experimental results of explainability on Normal Chest X-ray image using ResNet-18 Model through Grad-CAM++ visualization

The Grad-CAM++ results of ResNet-18 model has been illustrated in Figure 19 on normal chest X-ray image. The Figure 19 contains three panels first represents the original normal x-ray images m second shows Grad-CAM++ attention heatmap results with yellow, red and blue colors and on the right side shows the overlay of heatmap on original chest x-ray image and ensures that there is no central attention but it is wide spread across both sides of the lungs image, which confirms clinically that the input image exists in normal classification category.

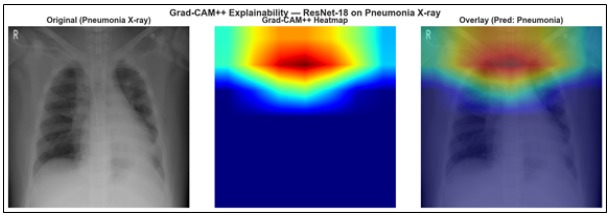


FIGURE 20: The experimental results of explainability on Pneumonia Chest X-ray image using ResNet-18 Model through Grad-CAM++ visualization

On the other hand, in Figure 20 shows the explainability results on pneumonia chest x-ray image using Pneumonia ResNet-18. From experimental results of Grad-CAM++ mentioned in the Figure 20 (Right) which confirms that model focus on pneumonia infected areas and aligned with radiological diagnosis practices, clinical validity, and interpretability as required in clinical decision-making process.

F. COMPARISON WITH STATE-OF-THE-ART RECENT STUDIES

We have considered the state and art comparison of our research with recent research studies conducted for pneumonia detection and considered the different research comparison criteria on various parameters as mentioned in the Table 9.

The comparison analysis in Table 9 clearly shows that our research has not only considered the accuracy but many other parameters which are very important for clinical deployment and use of deep learning models in the field of medical imaging. It also shows the existing research gap which our study has covered and contributed evidently. Furthermore, above comparison results show the importance, need, and novelty of this research study.

VI. CLINICAL SAFETY ANALYSIS

This research study critically highlights very useful multiple findings of two model architectures including the VGG-19 and Swin-Tiny face the sever problem of majority-class collapse in class prediction on slightly imbalanced images dataset. Both models are predicting 'Normal' for every image (F1 = 0.000, AUC = 0.500 per collapsed fold) in 3 out of 5 folds. Only accuracy does not guarantee the correct disease classification, meaning that a model with 95% specificity but having 0% sensitivity would fail to detect the pneumonia from x-ray images in real clinical practices. Based on the findings of the experimental results that despite of achieving high accuracy by both models in medical image analysis. It is highly recommended to be cautious before clinical deployment. In terms of real-world clinical perspective, the majority-class collapse in prediction would have serious clinical and diagnostic concern for proper diagnosis and patient's treatment if pneumoniae cases are misdirected through this

TABLE 9: Comparison analysis with recent research studies

Criterion	Pinem et al. (2026)	Oltu et al. (2025)	Aldabayan et al. (2026)	This Study
Dataset	3 CXR datasets	21,165 CXRs (single)	ChestX-ray14 (NIH)	31,540 (Kermany + RSNA)
Models Tested	13 (Transformers only)	ViT + DenseNet201	ViT only	10 (7 CNN + 3 ViT)
Best Accuracy	87.43% (5-model ensemble)	97.87%	70-75%	86.9%
Cross-Validation	No	No	No	5-fold stratified CV
Statistical Tests	No	No	No	McNemar's (45 pairs)
CNN vs. ViT	Transformers only	Hybrid only	ViT only	Full comparison (7 CNN, 3 ViT)
Explainability	No	No	No	Grad-CAM++ all 10 models
Efficiency Profile	No	No	No	FLOPs, latency, VRAM

issue. Therefore, we recommend that deep learning models' assessment and evaluation should be tested beyond accuracy of results and must consider other parameters including, sensitivity, fold-level stability, statistical significance, computational efficiency, and explainability analysis before model's selection and deployment in real-world clinical practices.

A. MAJORITY-CLASS COLLAPSE (CLINICAL SAFETY IMPLICATIONS)

This research suggests some strategies to deal with majority class collapse issue highlighted when we have imbalanced image dataset in the medical domain, which are summarized in Table 10.

TABLE 10: Suggested techniques for majority-class collapse in imbalanced datasets

Strategy	Implementation	Effectiveness
Stratified CV	Mandatory for all evaluations	High
Weighted loss	Weight = 1 / class frequency	Medium
Oversampling (SMOTE)	Generate synthetic minority samples	Medium (overfitting)
Undersampling	Reduce majority class size	Medium (data loss)
Ensemble methods	Combine stable model predictions	High

VII. CLINICAL DEPLOYMENT RECOMMENDATIONS

Based on the findings of this study, it is recommended that the deep learning models for clinical deployment and consideration of clinical context for pneumonia detection as mentioned in Table 11.

TABLE 11: Clinical deployment recommendations based on accuracy, efficiency, and safety

Deployment Scenario	Recommended Model	Key Rationale
Hospital-grade screening	EfficientNet-B0	Best accuracy (86.9%), lowest variance, AUC=0.932
Edge/mobile devices	MobileNetV3-Small	1.52M params, 45MB VRAM, 86.5% acc, 120 fps
Real-time emergency triage	ResNet-18	572.7 fps, sub-2ms inference, 86.4% acc
AVOID (clinically unsafe)	VGG-19 / Swin-Tiny	Majority-class collapse in 3/5 folds (F1→0)

VIII. RESEARCH LIMITATIONS

This study also highlights clear research limitations that need to be explored, tested with, and evaluated in future studies, as mentioned in Table 12.

TABLE 12: Research limitations and future strategies

Limitation	Potential Impact	Planned Mitigation
Single dataset combination	Results may not generalize to CheXpert/MIMIC-CXR	Validate on CheXpert (65,240) and MIMIC-CXR (377,110)
Binary classification only	No bacterial vs. viral discrimination	3-class extension (normal/bacterial/viral) ongoing
No COVID-19 specificity	Pandemic-era models need COVID-19 discrimination	COVIDx dataset under evaluation
Limited ViT hyperparameter search	ViT performance may improve with tuning	Grid search over patch size, LR, weight decay
Single GPU platform	Efficiency results may vary on CPU/edge devices	Intel Xeon CPU and Jetson Nano benchmarks planned

IX. CONCLUSION

This study has experimented and evaluated different deep learning architectures beyond the accuracy level. This proposed research framework evaluated the seven (7) Convolutional Neural Networks and three (3) Vision transformers for pneumonia classification and evaluating the performance of these ten (10) architectures in different deeper dimensions including accuracy, clinical safety, explainability analysis, statistical significance, 5-fold stratified cross-validation, and computation efficiency profiling. This benchmark study suggests and recommends statistically validated deep learning models for real-world deployment in clinical practices for chest x-ray images screening and diagnosis. This research study critically highlights very useful multiple findings with regards to two model architectures including the VGG-19 and Swin-Tiny which faced sever problem of majority-class collapse in class prediction on slightly imbalanced images dataset. Both models are predicting 'Normal' for every image (F1 = 0.000, AUC = 0.500 per collapsed fold) in 3 out of 5 folds. This study also ensures that dataset diversity is more crucial than its quantity. As we confirmed through experiments that our mixed source images dataset results more generalize models than single source pediatric images alone. This study also performs the 5-fold cross validation and pairwise testing through McNemar's test to measure the model's significant performance differences. The explainability experimental and visualization results through Grad-CAM++ approach ensures models success in clinical decision support and ensuring the radiologist trust and verification. The research study framework which is combining systematic comparison benchmarking, clinical safety analysis, comprehensive explainability, computational efficiency, statistical significance, and cross validation for evaluating deep learning models in high-stakes medical imaging appli-

cations.

X. FUTURE WORK

Based on the findings and limitations of this study, the future work can focus on the Multi-class pneumonia classification which can be extended to bacterial vs. viral vs. COVID-19 pneumonia (3 classes), Cross-dataset validation to evaluate SynthEnsemble on CheXpert (65,240 CXRs) and MIMIC-CXR (377,110 CXRs) to assess generalizability across institutions. The use of Lightweight ViT variants to compact vision transformers (e.g., MobileViT, TinyViT) may offer better efficiency than standard ViT-B/16. The federated learning for privacy-preserving multi-hospital training without centralized data sharing. The use of weakly supervised localization to extend binary classification to bounding box prediction and pneumonia localization. The Prospective clinical study to real-world evaluation in an emergency department setting to assess clinical utility and radiologist augmentation.

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